Mathematical and Computational Methods

Exercise Sheet 10: Integration II: techniques and applications

Tutorial route (▶, in this order, about 80 minutes): 1.2, 2.1(a,b), 2.5(a), 2.4(a), 2.8(a,c), 2.9(a,b), 2.10(a). Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Find the indefinite integral (with +C) unless limits are shown.

1.1 Techniques

Exercise 1.1 *(⋆ drill — by parts)*. (a) $\displaystyle\int x\,e^{2x}\,dx$, (b) $\displaystyle\int x\ln x\,dx$.

Exercise 1.2 *(▶ ⋆ drill — partial fractions)*. $\displaystyle\int \frac{3x+1}{x(x+1)}\,dx$.

Exercise 1.3 *(⋆ drill — trig integrals)*. (a) $\displaystyle\int \sin^2 x\,dx$, (b) $\displaystyle\int \cos^3 x\,dx$, (c) $\displaystyle\int \tan^2 x\,dx$.

Exercise 1.4 *(⋆ drill — trig substitution)*. $\displaystyle\int \frac{dx}{\sqrt{9-x^2}}$.

1.2 Applications

Exercise 1.5 *(⋆ drill — area between curves)*. Find the area enclosed between $y = 2x$ and $y = x^2$.

Exercise 1.6 *(⋆ drill — volume of revolution)*. The region under $y = \sqrt{x}$, $0 \le x \le 4$, is revolved about the x-axis. Find the volume.

Exercise 1.7 *(⋆ drill — average value)*. Find the average value of $f(x) = x^2$ on $[0, 3]$.

Exercise 1.8 *(⋆ drill — density)*. The random variable X has density $f(x) = 3x^2$ on $[0, 1]$ (and 0 elsewhere). Verify it is a density, then find $P(X > \frac{1}{2})$ and $\mathbb{E}[X]$.

2 Tutorial problems

Exercise 2.1 *(▶ ⋆⋆ core — by parts (worked-example anchor))*. Evaluate

$$\text{(a)}\ \int x^2 e^x\,dx, \quad \text{(b)}\ \int \arctan x\,dx, \quad \text{(c)}\ \int e^x \sin x\,dx.$$

Exercise 2.2 *(⋆⋆ core — partial fractions)*. Evaluate

$$\text{(a)}\ \int \frac{2x+3}{x(x+1)^2}\,dx, \quad \text{(b)}\ \int \frac{dx}{x^2+4x+13}, \quad \text{(c)}\ \int \frac{x^3}{x^2+1}\,dx.$$

Exercise 2.3 *(⋆⋆ core — trigonometric integrals).* Evaluate

$$\text{(a)} \int_0^{\pi/2} \sin^3 x \cos^2 x \, dx, \quad \text{(b)} \int_0^{\pi/4} \tan^3 x \, dx, \quad \text{(c)} \int \sec^4 x \, dx.$$

Exercise 2.4 *(▶ ⋆⋆ core — trigonometric substitution).* Evaluate

$$\text{(a)} \int \sqrt{9 - x^2} \, dx, \qquad \text{(b)} \int \frac{dx}{(x^2 + 4)^{3/2}}.$$

Exercise 2.5 *(▶ ⋆⋆ core — area between curves (worked-example anchor)).*

(a) Sketch the region first, then find the area enclosed between the parabola $y = x^2$ and the line $y = x + 2$.

(b) Find the total area between $y = \sin x$ and $y = \cos x$ for $0 \le x \le \frac{\pi}{2}$.

Exercise 2.6 *(⋆⋆ core — volumes of revolution).*

(a) Find the volume when the region under $y = \sin x$, $0 \le x \le \pi$, is revolved about the x-axis.

(b) Revolving $y = \sqrt{r^2 - x^2}$, $-r \le x \le r$, about the x-axis sweeps out a sphere of radius r. The *cap* of height h ($0 < h \le 2r$) is the part with $r - h \le x \le r$. Show that its volume is $V = \frac{1}{3}\pi h^2(3r - h)$, and check that $h = 2r$ gives the whole sphere.

(c) *Spot the error.* For the region of Warm-up 1.6 a student writes $V = \pi \int_0^4 \sqrt{x} \, dx = \pi \left[\frac{2}{3}x^{3/2}\right]_0^4 = \frac{16\pi}{3}$, instead of the 8π found there. What went wrong, and what is the correct set-up?

Exercise 2.7 *(⋆⋆ core — improper integrals).* Evaluate, or show divergent:

$$\text{(a)} \int_0^\infty e^{-2x} \, dx, \quad \text{(b)} \int_0^\infty x e^{-x^2} \, dx, \quad \text{(c)} \int_1^\infty \frac{dx}{x^p} \ (p > 0).$$

Exercise 2.8 *(▶ ⋆⋆ core — probability and expectation (data science)).* *(Exponential law.)* The lifetime T (hours) of a component has density $f(t) = \lambda e^{-\lambda t}$ on $[0, \infty)$ with $\lambda = \frac{1}{2}$.

(a) Find $P(T > 2)$.

(b) Find $P(1 < T < 3)$.

(c) Find the mean lifetime $\mathbb{E}[T]$.

Exercise 2.9 *(▶ ⋆⋆ core — setting up (physics, work)).* A uniform rope of length $L = 10\,\text{m}$ and mass $\mu = 2\,\text{kg}$ per metre hangs from a winch at the top of a building. Take $g = 9.8\,\text{m}\,\text{s}^{-2}$.

(a) Find the work done to wind the whole rope to the top.

(b) Find the average depth $\bar{x}$ below the top of the points of the hanging rope, and show that the work in (a) equals the rope's weight $\mu g L$ times $\bar{x}$.

Exercise 2.10 *(▶ ⋆⋆ core — volume by cross-sections).*

(a) A pyramid of height h has as its base an equilateral triangle of side a. Measuring x down from the apex, use similar triangles to find the area $A(x)$ of the cross-section at depth x, and show that $V = \frac{\sqrt{3}}{12}a^2 h$.

(b) ⋆⋆⋆ challenge A solid has a circular base of radius R in the xy-plane. Every cross-section perpendicular to a fixed diameter is a square. Find its volume.

3 Further practice (home)

3.1 Techniques

Exercise 3.1 *(★★ core — linear over quadratic).* $\displaystyle\int \frac{2x+3}{x^2+9}\,\mathrm{d}x.$

Exercise 3.2 *(★★ core — trig substitution (secant)).* $\displaystyle\int \frac{\mathrm{d}x}{\sqrt{x^2-9}}$ (for $x>3$).

3.2 Applications

Exercise 3.3 *(★★ core — average value (physics)).* A particle moves with velocity $v(t) = 6t - t^2$ ($\mathrm{m\,s^{-1}}$) for $0 \le t \le 6$. Find its average velocity over this interval.

Exercise 3.4 *(★★ core — find the constant).* Find k so that $f(x) = kx$ is a probability density on $[0, 4]$, then compute $\mathbb{E}[X]$.

Exercise 3.5 *(★ drill — setting up).*

(a) A rod occupies $0 \le x \le L$ with linear density $\rho(x) = kx$. Find its mass.

(b) *(Physics.)* A spring of stiffness $200\,\mathrm{N\,m^{-1}}$ needs the force $F(x) = 200x$ to hold it stretched by x metres. Find the work done in stretching it from its natural length to $x = 0.1\,\mathrm{m}$.

3.3 Conceptual

Exercise 3.6 *(★★ core — true or false?).* Decide, with a one-line reason, whether each statement is true or false.

(a) Every rational function has an elementary antiderivative (after polynomial division, if needed).

(b) $\displaystyle\int_{-\infty}^{\infty} x\,\mathrm{d}x = 0$ because the integrand is odd.

(c) The average value of a continuous f on $[a, b]$ always lies between the minimum and maximum of f on $[a, b]$.

3.4 Challenges and extensions

Exercise 3.7 *(★ ★ ★ challenge — mass and centre of mass).* A rod on $0 \le x \le L$ has density $\rho(x) = \rho_0\big(1 + \frac{x^2}{L^2}\big)$. Find its mass M and the position $\bar{x} = \dfrac{1}{M}\displaystyle\int_0^L x\,\rho(x)\,\mathrm{d}x$ of its centre of mass.

Exercise 3.8 *(★ ★ ★ challenge — normalising the Gaussian (data science)).* Given the standard result $\displaystyle\int_{-\infty}^{\infty} \mathrm{e}^{-x^2}\,\mathrm{d}x = \sqrt{\pi}$, find the constant c making $f(x) = c\,\mathrm{e}^{-x^2}$ a probability density on $\mathbb{R}$, and find $\mathbb{E}[X]$.